Business Process Prediction, Simulation and Optimization

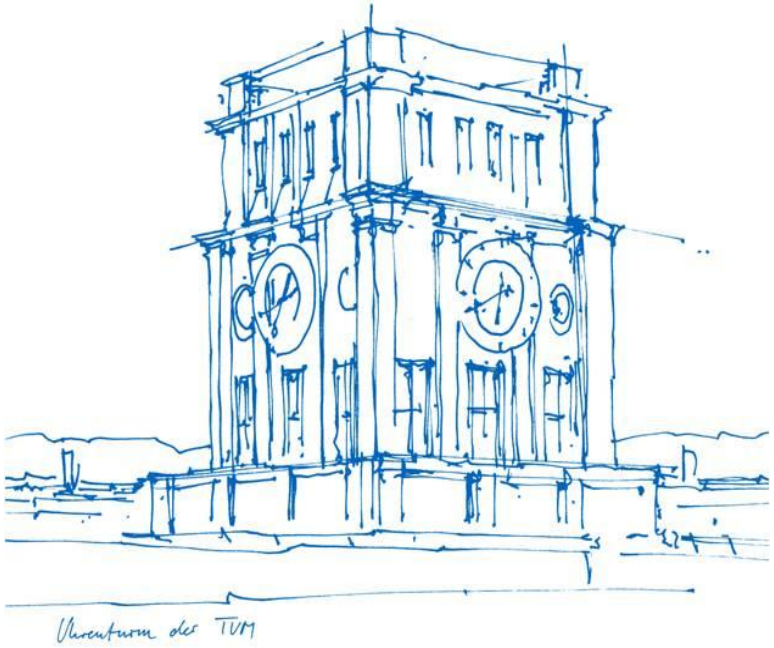
1. Introduction

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Technical University of Munich

Garching, 15 October 2026



Course Organization



Henryk Mustroph

Course Overview

Motivating example

“Our process is too inefficient!

We have:

1. High costs
2. Low revenue

What should we do?”



Course Overview

Interviews and process science





Process Science

Course Overview

This course covers prediction, simulation, and optimization of a loan application process at a financial institution.



BPIC-17 Event Log

Source: https://commons.wikimedia.org/wiki/File:2020_ING_Bijlmerdreef_106_(1).jpg

Overall Course Goal

Three steps from event log to optimization policies

1

Receive an event log, design a process model first

2

Design a data-driven business process simulation model

3

Propose optimization policies based on the simulation model

Organizational

Timeline

Two phases over the semester

Phase I

- Lectures
- Individual work on process mining
- Group finding
- Assignment I

Phase II

- Lectures
- Group work on simulation and optimization
- Assignment II
- Final presentation

Timeline

Two phases over the semester

Phase I

- Lectures
- Individual work on process mining
- Group finding
- Assignment I: Individual report

Phase II

- Lectures
- Group work on simulation and optimization
- Assignment II
- Final presentation

Phase I: Lectures

Topics of the lecture sessions

Business Processes (this session)

- Timeline and semester plan
- Business Process Management introduction
- From Workflow Engines to Process Mining
- Research timeline and companies

Process Mining

- Data standards and requirements
- Process mining with Python: PM4Py
- Process discovery
- Conformance checking
- Decision mining

Phase I: Assignment I

Individual work on process mining

- You will work with different Python libraries to analyze the event log and perform process mining, i.e., process discovery, conformance checking, and decision mining (Master only).
- You will create, evaluate, iteratively refine, and finalize a process model from the real-world BPIC-17 event log.

> You have to write and submit a report in LaTeX (max. 5 pages).

Phase I: Goal

Goal of Assignment I

> Based on the real-world event log, **identify** all important insights, reflect on and **discuss** them, and **design**, **evaluate**, and **present** a process model that visualizes how the process runs.

Timeline

Two phases over the semester

Phase I

- Lectures
- Individual work on process mining
- Group finding
- Assignment I: Individual report

Phase II

- Lectures
- Group work on simulation and optimization
- Assignment II
- Final presentation

Phase II: Lectures

Topics of the lecture sessions

Business Process Simulation

- Introduction to data-driven simulation
- Simulation engine
- Simulator components

Predictive Process Monitoring

- Machine learning and deep learning basics
- Data and predictive models for process monitoring
- Using predictive process monitoring models in simulation

Business Process Optimization

- Resource optimization through simulation
- Process improvements

Phase II: Assignment II

Group work on simulation and optimization

Assembling a data-driven business process simulation model

- Four course participants in each group
- Each group member works on a separated part of the simulation model
- Typically, everyone works on a different process perspective

Implementing and evaluating optimization policies

- Implement different policies
- Strategies to evaluate their impact
- Answer management questions based on your simulation

You have to write and submit a report in LaTeX (max. 10 pages) and give a final presentation as a group (20 min).

Phase II: Goal

Goal of Assignment II

> Based on the real-world event log, your group **designs** a business process simulation model and **implements** it as a digital twin of the past process. The simulator enables what-if analyses and process forecasts, and serves as a test environment for process improvements and prescriptive actions.

General Requirements

What the course expects from you



You will need to **work independently**, based on concepts from the scientific literature



You will need **strong Python programming skills** to implement theoretical concepts



You will need some knowledge of **statistics, machine learning, and optimization techniques**



You will need to spend **considerable time** on the course between October and February

Course Goal

Individual assignment and group work

Individual assignment I (25% of the total grade)

- Investigate and analyze the event log BPIC-17
- Perform basic data science and process mining to gain insights
- **Your grade is based on the report.**

Group assignment II (75% of the total grade)

- Design a data-driven business process simulation model
- Perform business process optimization through simulation
- **You will be graded individually based on your part in the report, your code, and your part in the presentation.**

Meetings

Course sessions on Thursdays

Practical Course (Bachelor's)	Advanced Practical Course (Master's)
Thursdays, 13:30 (90 min)	Thursdays, 15:00 (90 min)

Meetings: Preliminary Schedule

Attendance is mandatory

15 October	Introduction to the course and to Assignment I
22 October	Lecture on Process Mining I
29 October	Lecture on Process Mining II

Meetings: Preliminary Schedule

Attendance is mandatory

5 November	Q&A session
12 November	Guest lecture on SAP Signavio: introduction and workshop on the SAP Signavio Process Mining Suite
19 November	Submission Assignment I; Lecture on Business Process Simulation; introduction to Assignment II; group formation
26 November	Lecture on Machine Learning and Predictive Process Monitoring

Meetings: Preliminary Schedule

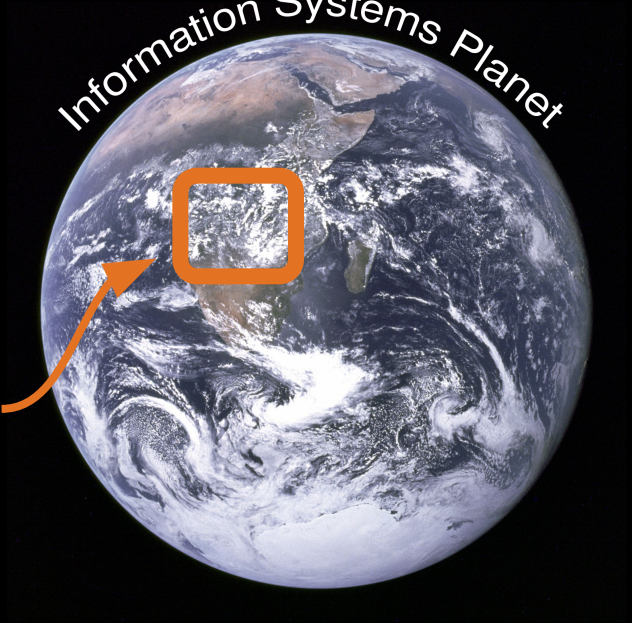
Attendance is mandatory

3, 10, 17 December	Q&A sessions
7 January	Lecture on Optimization
14, 21 January	Q&A sessions
28 January	Q&A session; information on the final presentation (mandatory for at least one group member)
4 February	Submission of Assignment II and final presentation

Let’s get started…

Where are we?

Course topics in the context of BPM



Business Process Management

What is BPM?

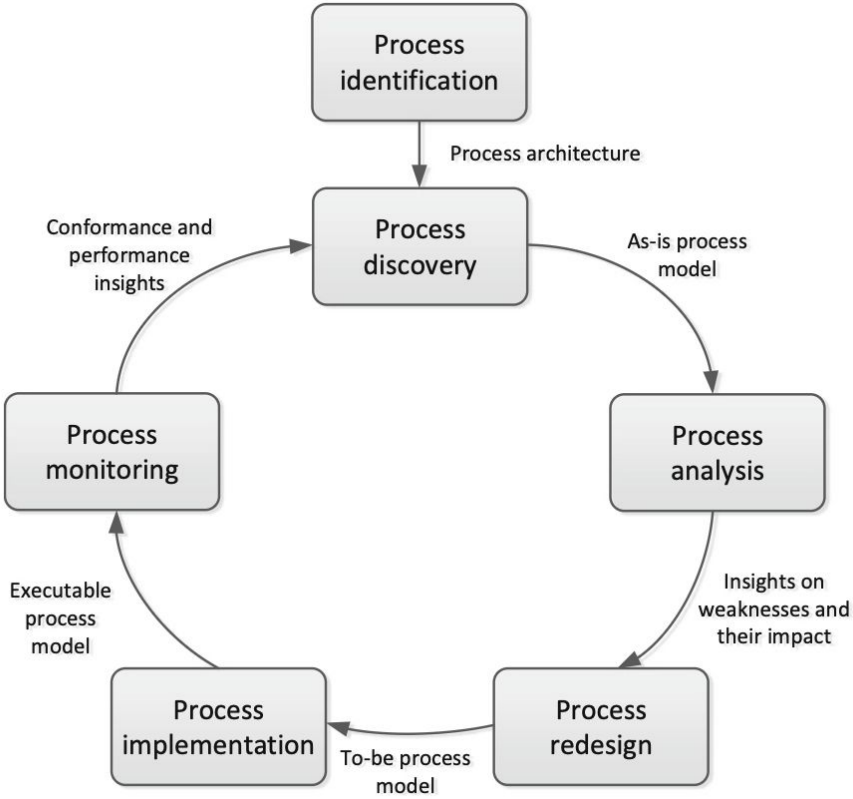
Definition

> *“Business Process Management (BPM) is the art and science of overseeing how work is performed in an organization to ensure consistent outcomes and to take advantage of improvement opportunities.” [1]*

What is BPM?

Lifecycle

BPM is a lifecycle of interconnected tasks [1].



Source: [1]

Business Processes

Business Process

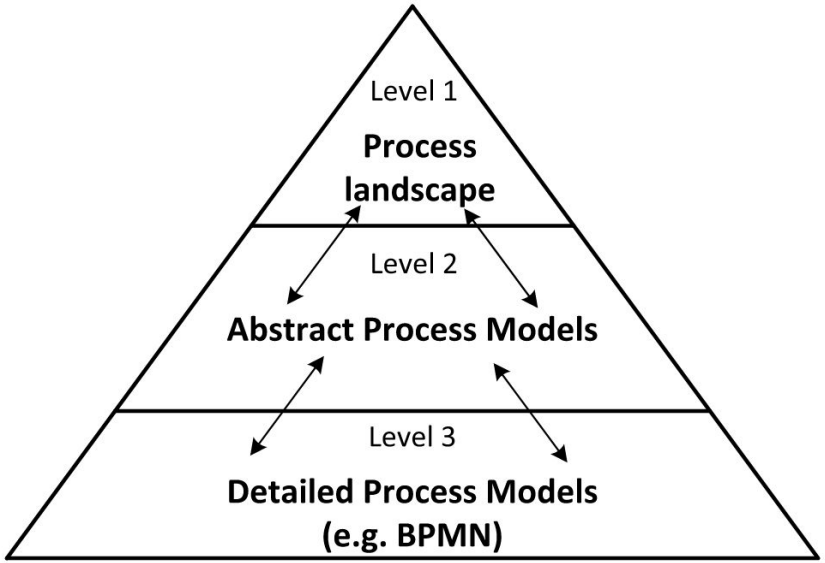
Definition and components

> *"[A business process is] a collection of interrelated events, activities, and decision points that involve a number of actors and objects, which collectively lead to an outcome that is of value to at least one customer." [1]*

Events	Happen automatically and have no duration
Activities	Take time; tasks are simple activities, seen as one unit of work
Decision points	Moments when a decision affects how the process is executed
Resources (actors)	Human actors, organizations, or software systems
Objects	Physical (equipment, materials, …) or immaterial (electronic records, …)

Classification of Business Processes

Overview



Source: [1]

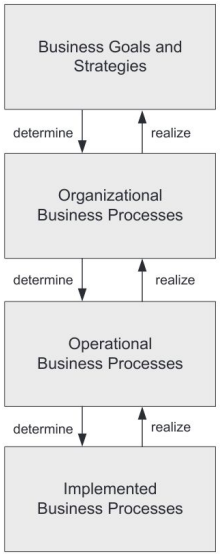
Classification of Business Processes

Degree of structuring

Highly structured *aka Lasagna process [2]*	• All activities are repeatable, with well-defined input and output [2] • Model prescribes activities and constraints completely [3] • Can hinder experienced, competent participants [3] • Danger: alignment
Semi-structured	• Information requirements of activities are known [2] • Execution can be sketched, but users may deviate from it [2]
Unstructured *aka Spaghetti process [2]*	• Driven by experience [2] • Rules of thumb, … [2]

Classification of Business Processes

Levels of abstraction



Organizational business processes

Typically specified in textual form by their inputs, outputs, expected results, and dependencies on other organizational business processes

- Informal or semi-formal (optimization) techniques

Operational business processes

Specified by business process models

- Often a 1:n relationship with organizational business processes

Implemented business processes

Contain information on the execution of the process activities

Source: [3]

Classification of Business Processes

Degree of repetition

Highly repetitive processes

- E.g., online airline ticketing
- Often, but not always, fully automated (e.g., insurance claims)
- Investments in optimizing the process can pay off

Infrequent processes

- Large engineering efforts, …
- Questionable whether process modeling, automation, or optimization pays off

Source: [3]

Classification of Business Processes

Degree of automation

Fully automated processes	Partly automated processes	Manual processes
● No humans are involved in the enactment	● Mixture of manual and automated activities	● Purely manual activities

Source: [3]

Classification of Business Processes

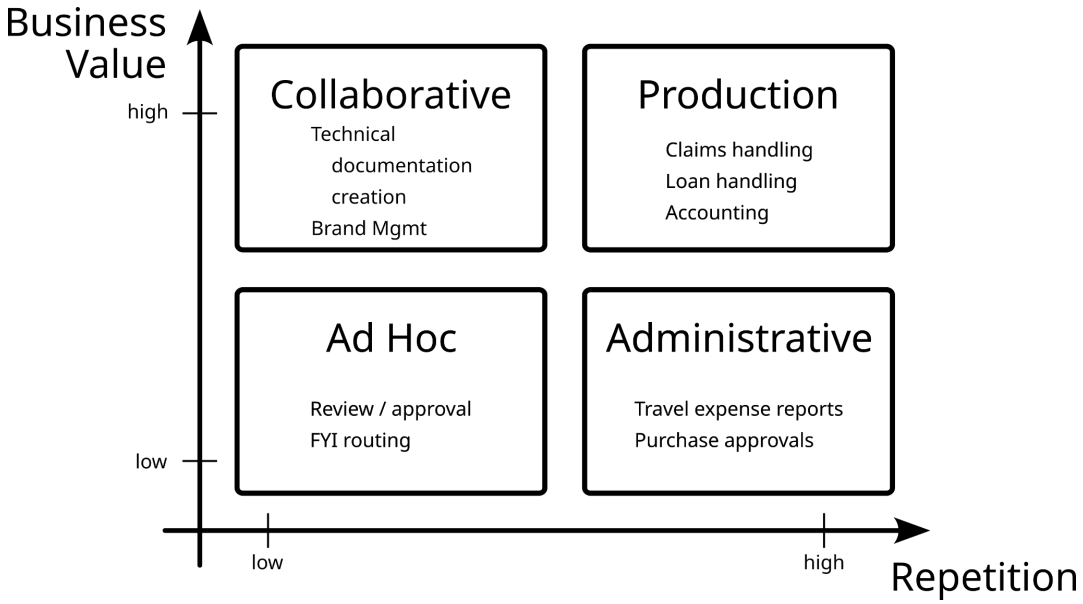
Involved organizations

Intra-organizational processes	Process choreographies
● No interaction with business processes of other parties	● Interactions of business processes across organizations ● **Danger:** communication aspects, legal matters, heterogeneous software infrastructure, …

Source: [3]

Classification of Business Processes

Collaborative, ad hoc, administrative, and production processes



Source: [4]

Process Selection

Which processes should we optimize quantitatively?

- Not too abstract processes:
  - Operational / implemented business processes
- Highly repetitive processes
- Business value (?)
- Degree of automation (?)
- Intra-organizational processes vs. process choreographies (?)
- Structured vs. unstructured processes (?)

Excursus

History of BPM and relevant companies

History to Future of BPM

From process reengineering to agentic BPM

1990s — **Process reengineering and workflow systems:** business process reengineering (BPR), formal workflow management systems, focus on efficiency and removing handoffs.

2000s — **BPM suites and standards:** centralized BPM platforms, standards for modeling (e.g., BPMN), and integration with enterprise systems. Emphasis on executable models and process governance.

2010s — **Monitoring, RPA, and process mining emerge:** operational logs became mainstream; process mining (log-based discovery, conformance checking, enhancement) matured; RPA automated repetitive GUI tasks.

2020s — **Data-driven and AI:** predictive monitoring, prescriptive analytics, ML for anomaly detection and resource allocation; orchestration of humans and bots; more real-time, event-stream processing.

Today — **Agenti BPM:** agentic process modelling, execution, prescription, …

From Workflow Engines to Process Mining

Workflow engines

Automate your process model

- Each activity can be seen as a small service
- Create a worklist that defines which resources execute which activities
- Orchestrate the control flow with the engine
- Start parallel executions (instances)
- Capture the data flow and store execution data in event logs

Workflow engine of our chair



https://cpee.org/

Advantages? Disadvantages?

From Workflow Engines to Process Mining

Industry adoption

However, industry did not adopt workflow engines: the research hype of the 1990s did not translate into practical use.

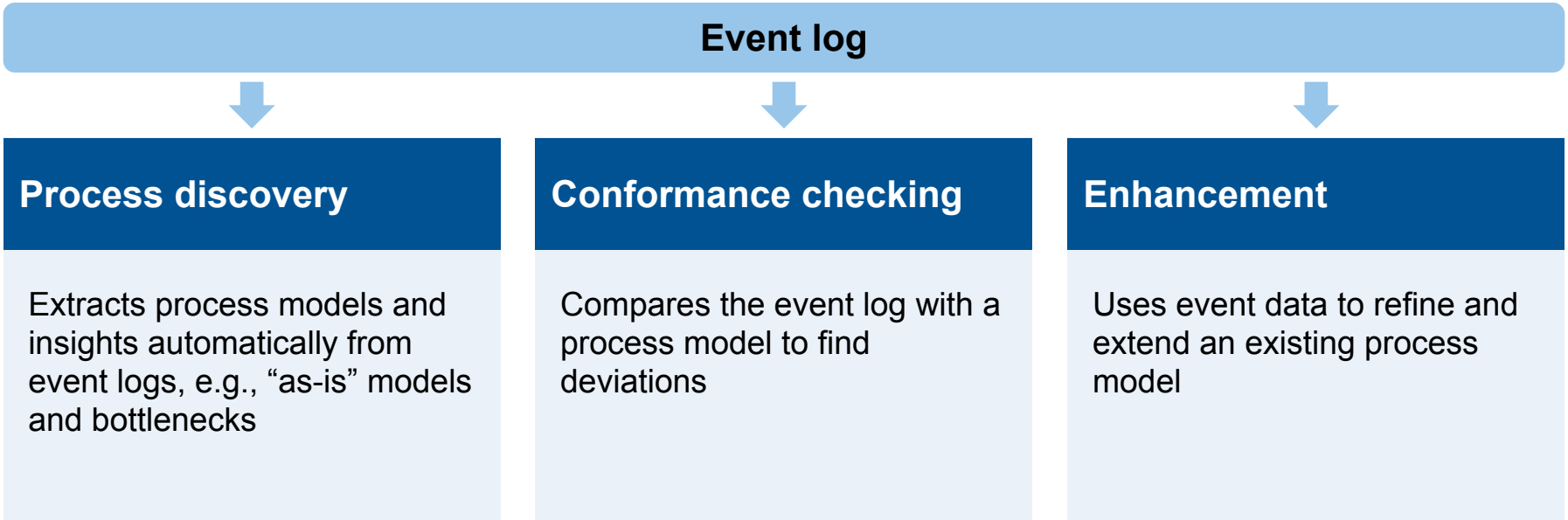
Why?

Podcast recommendation: “Godfather des Process Mining – Wil van der Aalst” (Die Prozess-Philosophen, episode 54)

https://die-prozess-philosophen.podigee.io/54-godfather-des-process-mining-wil-van-der-aalst

Process Mining

Process discovery, conformance checking, and enhancement



Source: [2]

Process Mining: Companies

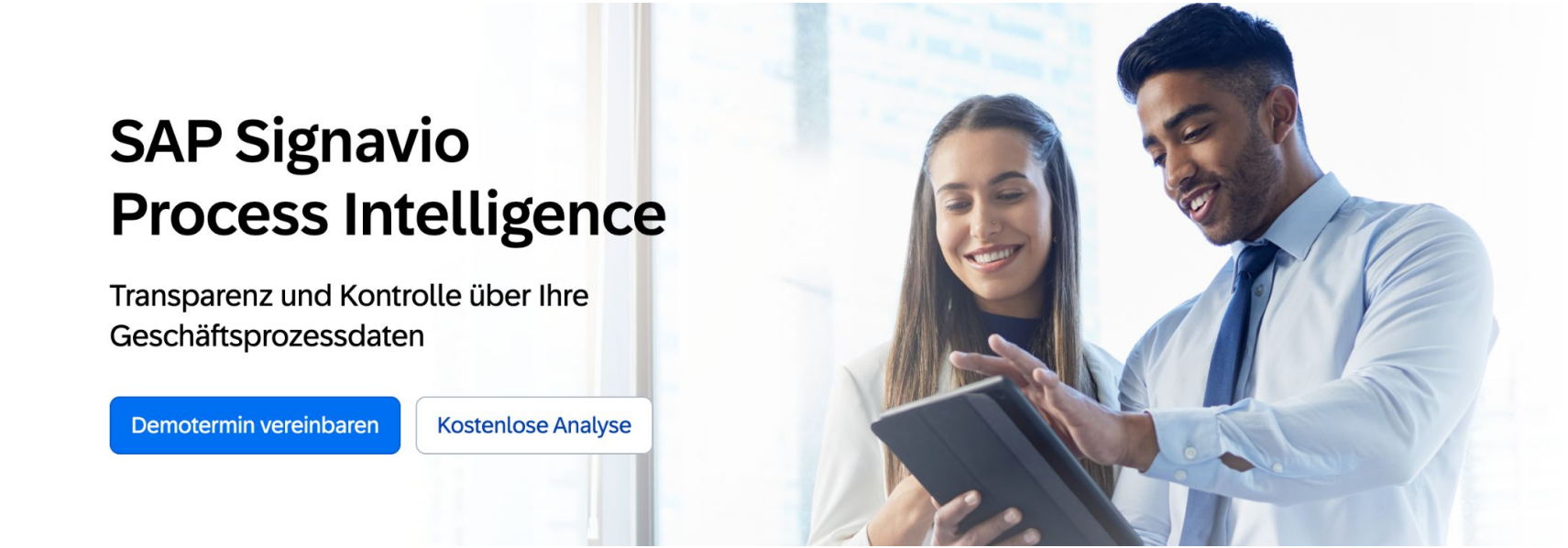
Selected vendors

Figure 1: Magic Quadrant for Process Mining Platforms



SAP Signavio

Process Intelligence



Source: https://www.signavio.com/de/products/process-intelligence/

Celonis

Process Intelligence Platform



Source: https://www.celonis.com/de

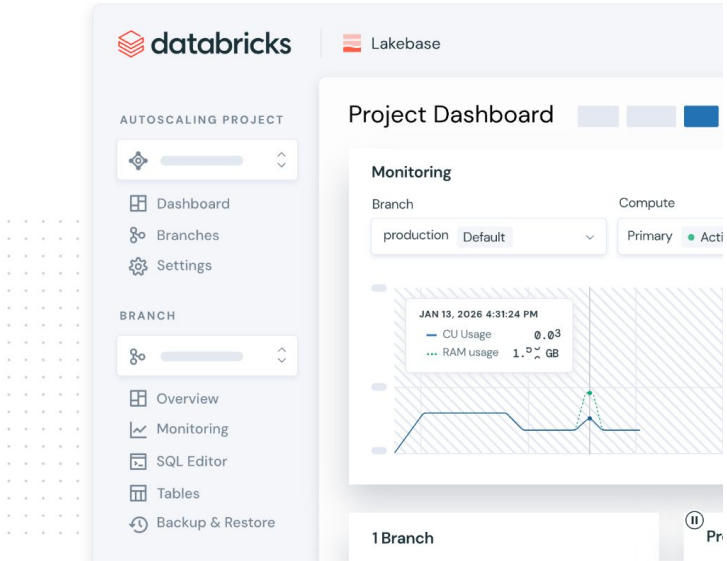
Other (US) Competitors?

Data and analytics platforms

Data platform	Intelligent (process) analytics platform
Databricks	**Palantir**

Databricks

Data platform



Source: https://www.databricks.com/

Databricks

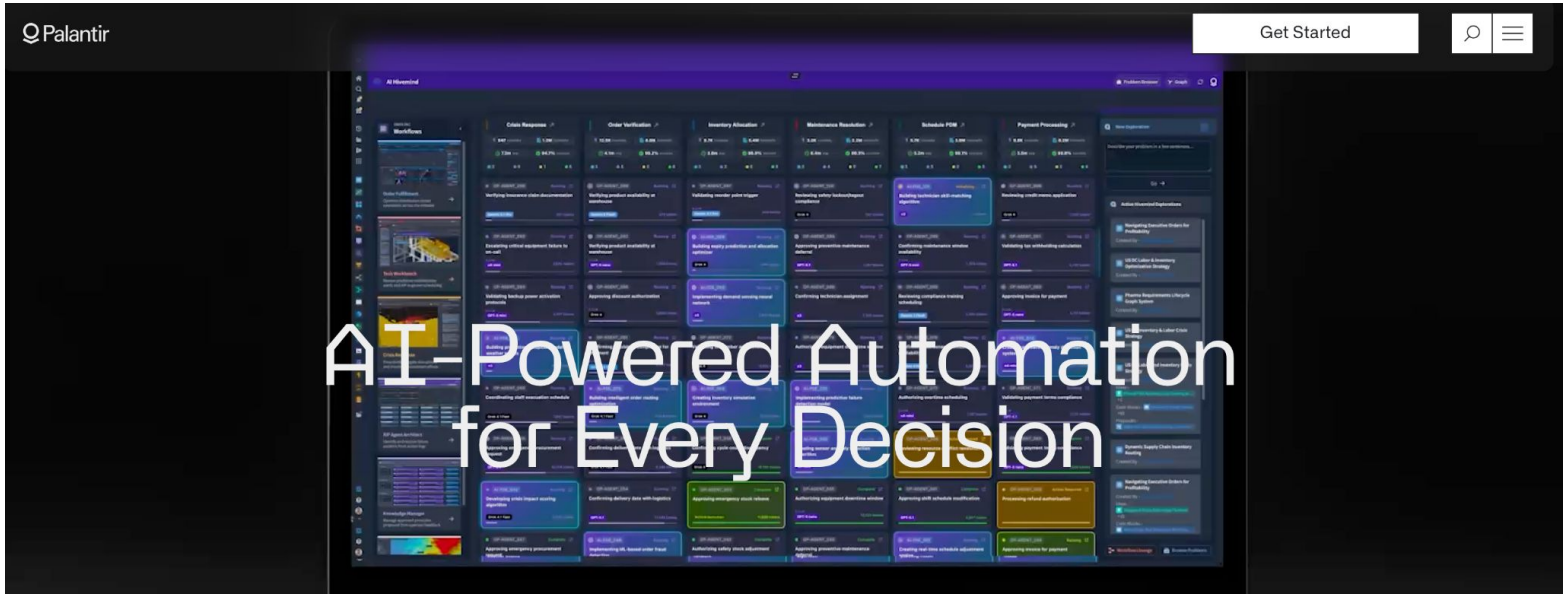
Data platform compared to Celonis

	Celonis	**Databricks**
Data connection	200+ prebuilt connectors	Lakeflow Connect
Data storage	Celonis Data Core	Lakehouse on Delta Lake
Data preparation	Prebuilt event models	Spark and SQL pipelines
Data aggregation	PQL on process data	SQL and Spark aggregations

Sources: Celonis press release (4 November 2025); product documentation

Palantir

Intelligent analytics platform



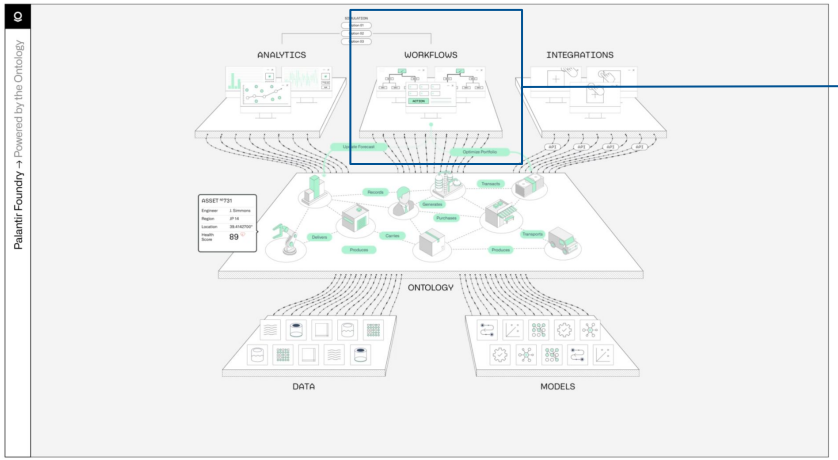
Source: https://www.palantir.com/

Palantir

A Celonis competitor beyond process mining

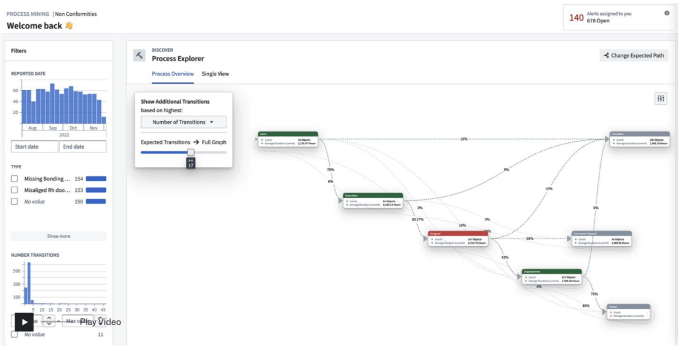
Palantir Foundry

The Ontology-Powered Operating System for the Modern Enterprise



Foundry Process Mining & Automation

Mine processes. Reveal inefficiencies. Drive change. Continuously optimize.



Source: https://www.palantir.com/platforms/foundry/

History to Future of BPM

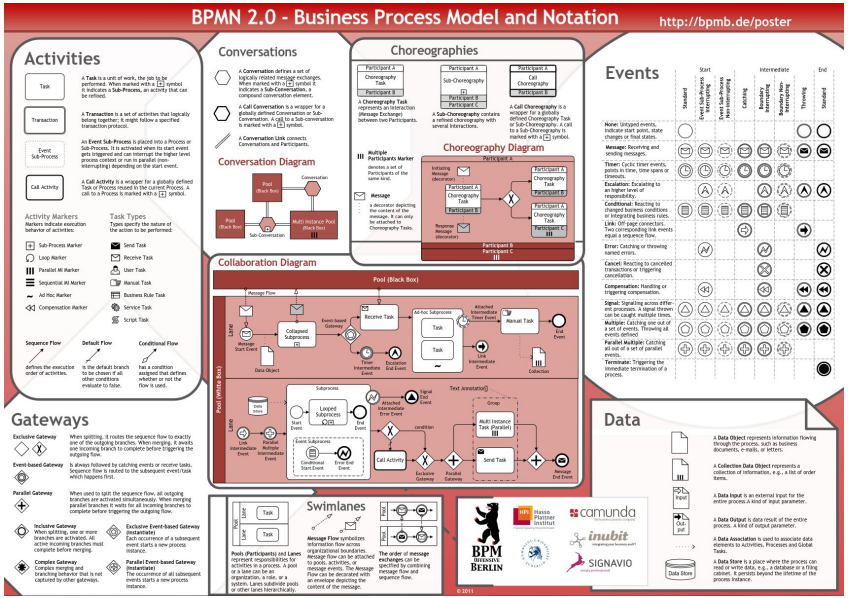
Discussion

> What do you think will come in the future?

Process Modeling

Business Process Model and Notation

BPMN



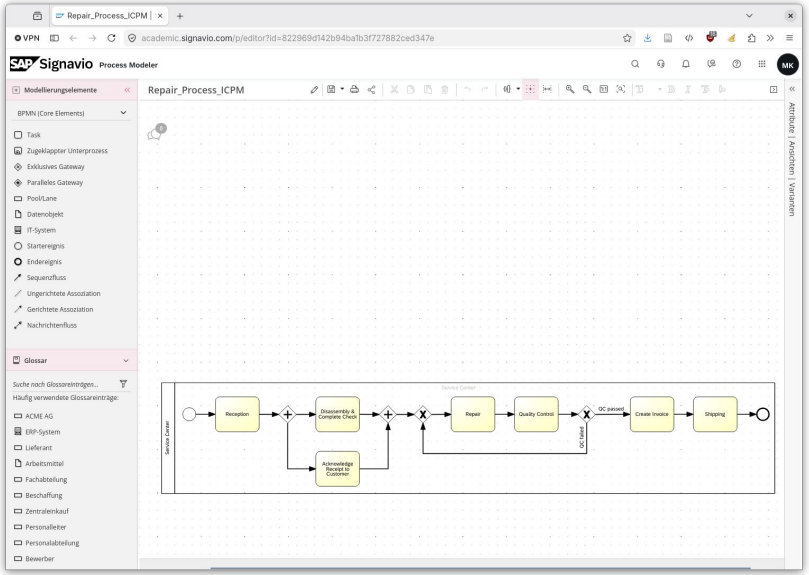
Source: https://bpmd.de/poster

Signavio

Academic account

Make your own student account:

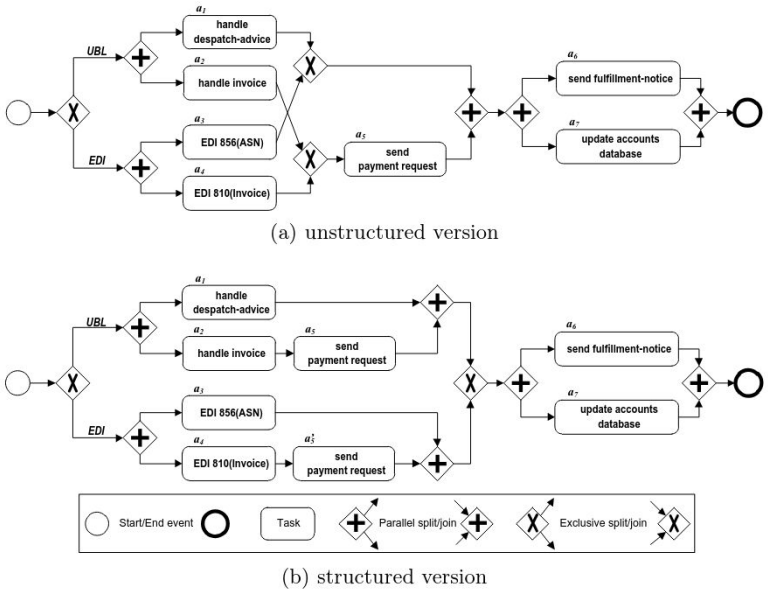
https://academic.signavio.com/



Block-Structured Process Models

Definition and limitation

For every node with multiple outgoing arcs (a **split**) there is a corresponding node with multiple incoming arcs (a join), such that the subgraph between the split and the join forms a **single-entry-single-exit (SESE)** region.



(a) unstructured version

(b) structured version

> Not all unstructured process models can be transformed into equivalent structured models.

Source: [5]

Literature

References

[1] Dumas, M., La Rosa, M., Mendling, J., & Reijers, H. A. (2018). *Fundamentals of Business Process Management* (2nd ed.). Springer.

[2] van der Aalst, W. (2016). *Process Mining: Data Science in Action* (2nd ed.). Springer.

[3] Weske, M. (2019). *Business Process Management: Concepts, Languages, Architectures* (3rd ed.). Springer.

[4] Leymann, F., & Roller, D. (2000). *Production Workflow: Concepts and Techniques*. Prentice Hall.

[5] Dumas, M., La Rosa, M., Mendling, J., Mäesalu, R., Reijers, H. A., & Semenenko, N. (2012). Understanding business process models: The costs and benefits of structuredness. In *Advanced Information Systems Engineering (CAiSE 2012)*, LNCS 7328 (pp. 31–46). Springer.